Examiner only

5. Temporary hard water is caused by the presence of dissolved calcium hydrogencarbonate, $Ca(HCO_3)_2$.

The table shows three different methods of removing temporary hardness from water.

Method	Product(s) in softened water
1. Adding sodium carbonate	insoluble calcium carbonate dissolved sodium hydrogencarbonate
2. Boiling	insoluble calcium carbonate carbon dioxide
3. Passing through an ion exchange column containing Na^+ ions	dissolved sodium hydrogencarbonate

(a) Complete the symbol equation for the reaction taking place in Method **1**. [1]

$$Ca(HCO_3)_2 \ + \ \ldots\ldots\ldots\ldots\ldots \longrightarrow CaCO_3 \ + \ 2NaHCO_3$$

(b) Underline the ratio of calcium ions to sodium ions exchanged in Method **3**. [1]

2:1 **1:1** **1:2** **2:2**

(c) Give the number of the method which does **not** form limescale. Give the reason for your answer. [2]

Number ....

Reason ....

....

(d) Tick (✓) the box next to the name of a compound that causes permanent hardness when dissolved in water. [1]

calcium sulfate ☐

potassium sulfate ☐

magnesium hydrogencarbonate ☐

sodium sulfate ☐

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